Survey of Materials

Introduction

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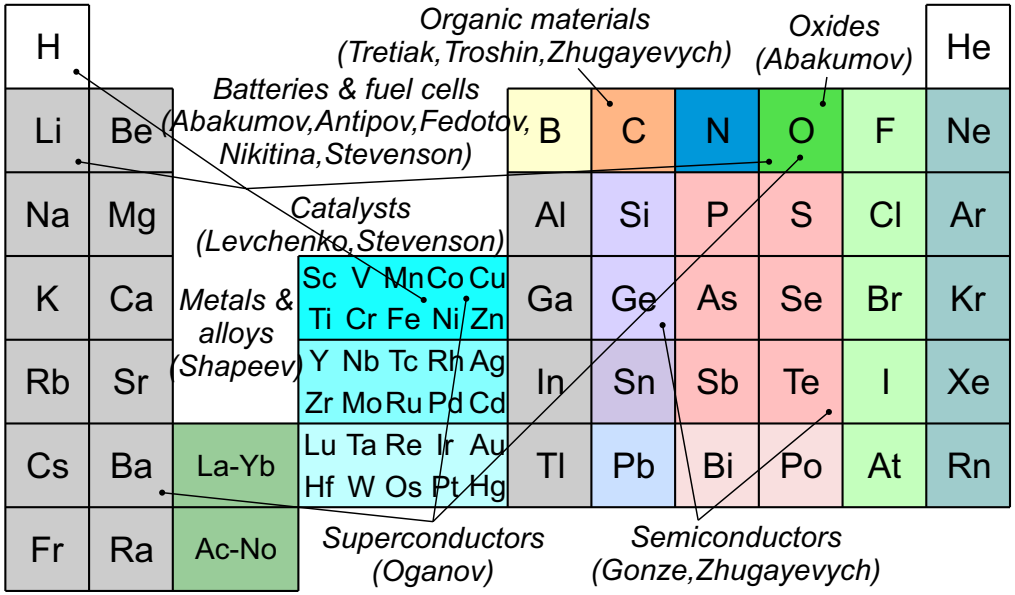
Outline

- What is this course about
- Case study 1: solar cells – energy generation
- Case study 2,3: batteries – energy storage
- Part II overview
- Course logistics

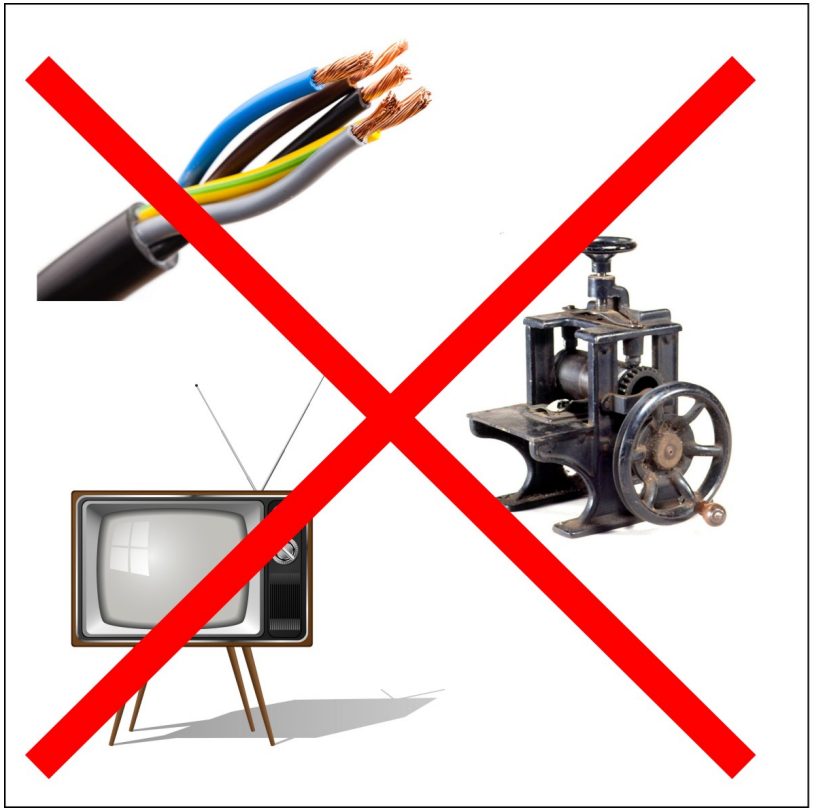
What is this course about

- Part I (1 week): Fundamentals of Materials Science to be able to understand Part II
- Part II (2 weeks): A set of independent lectures about materials given by experts in the corresponding field



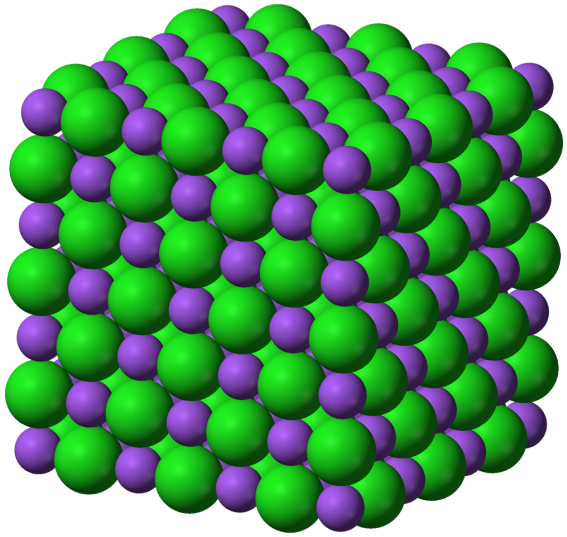
Also: Perovskites for optoelectronics (Tretiak,Troshin), Carbon nanomaterials (Nasibulin), Materials at high-pressure (Oganov), Hierarchically structured materials (Korsunksy)

What materials are covered



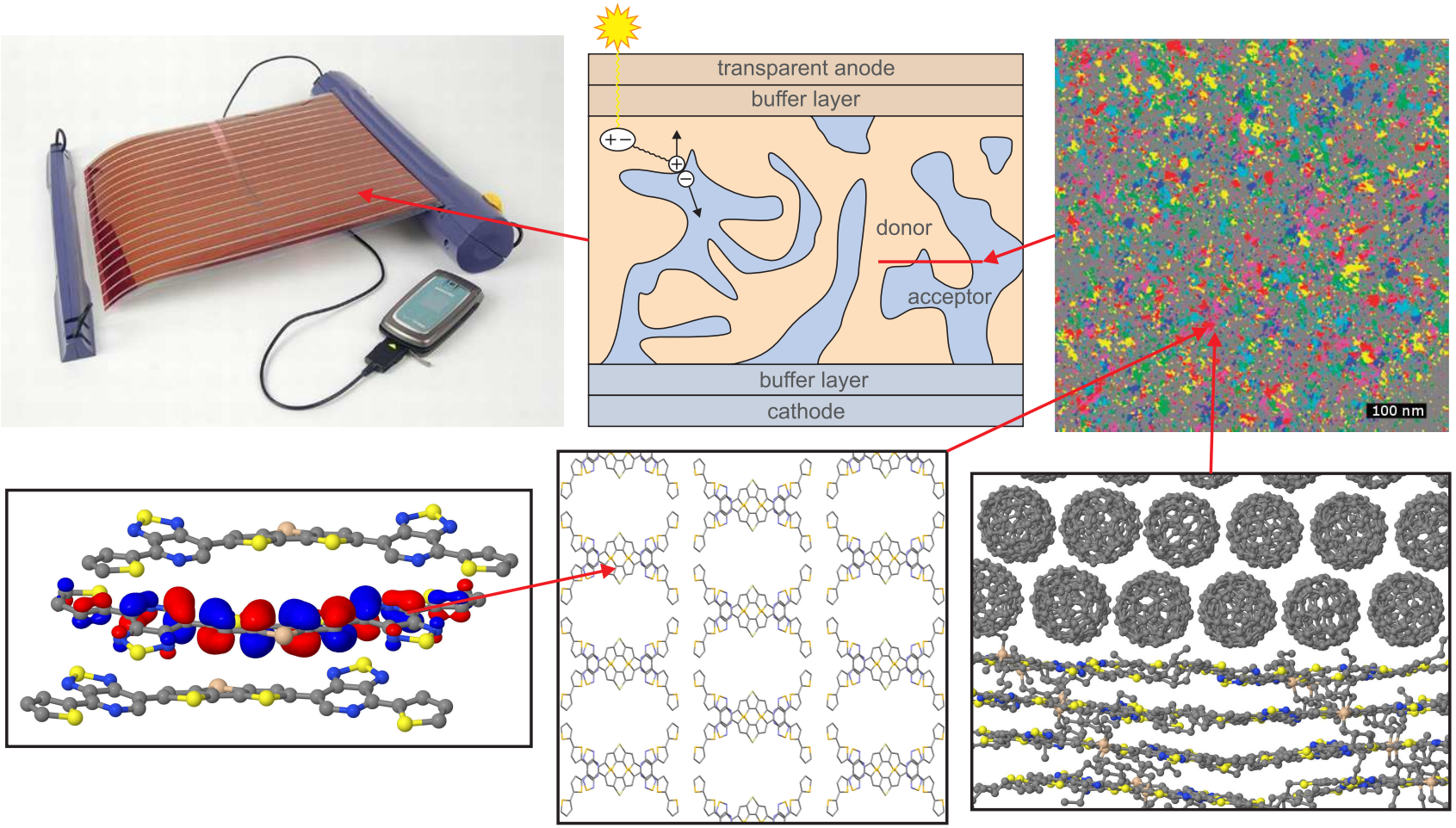
- New materials or new technologies
- Under research at Skoltech

What scales are covered



- Materials Science starting from microscopic scale
- Don't forget about other scales (meso-, macro-, device) and Materials Engineering and Technology

What scales are covered



Level of coverage

Basics + Overview + Readiness to special courses in Materials Science:

- *General courses*
  - Materials Chemistry
  - Materials Physics (Introduction to Solid State Physics)
  - Materials Engineering (Applied Materials and Design)
- *Modeling*
  - Materials Modeling (& Computational Chemistry + Advanced)
  - Structure and Property of Materials
- *Characterization*
  - Materials Structure Characterization
  - Electrochemistry
- *Specific materials*
  - (Electrochemical Energy Storage Materials)
  - Organic Materials
  - Carbon Nanomaterials

See roadmap here:
crei.skoltech.ru/cest/education/materials-science-program

Addressing regional challenges in high-tech manufacturing

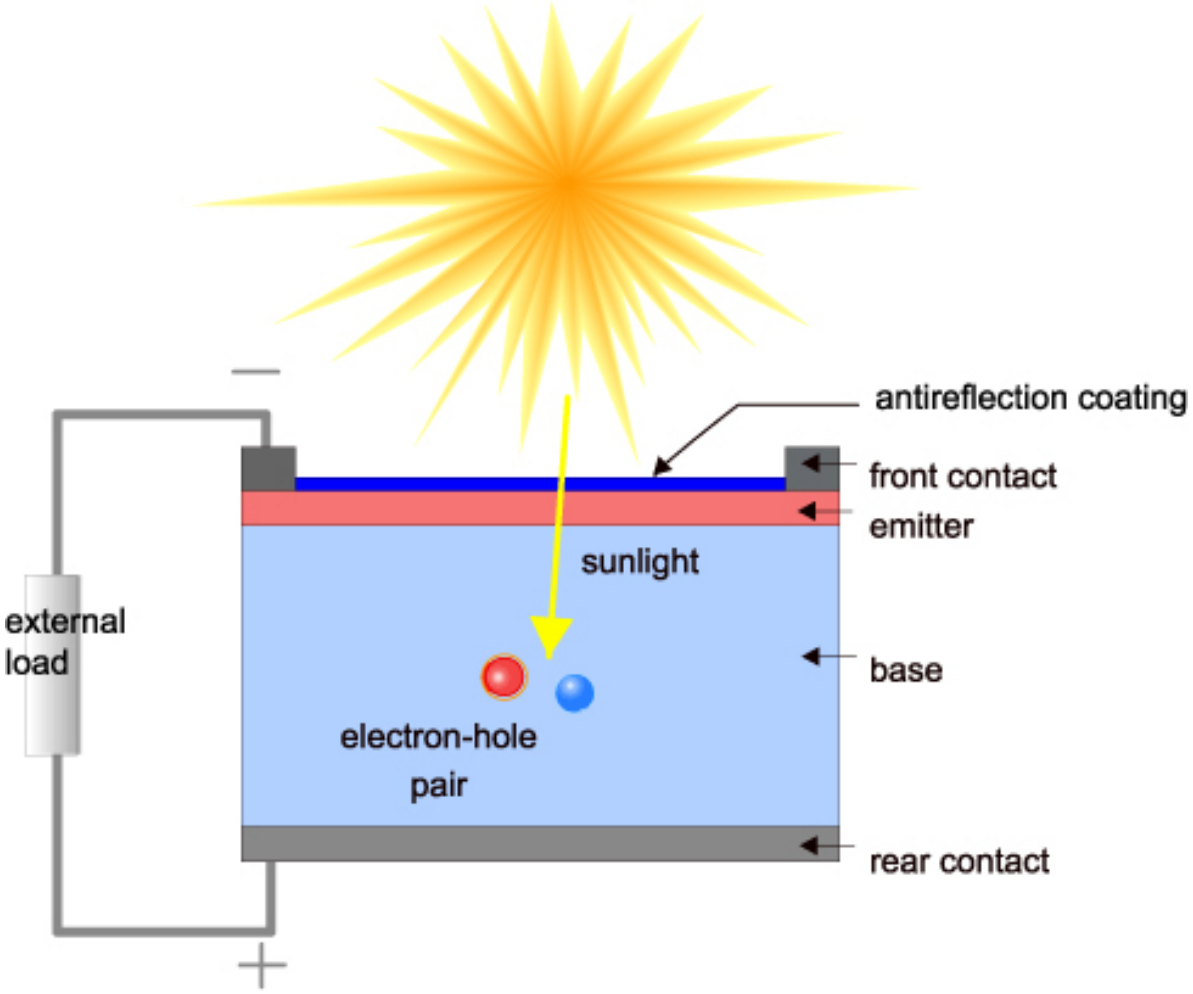
Where is the bottleneck?

- feedstock
- materials
- parts
- assembly
- software
- sales
- service

MSE addresses materials + parts

Case study 1: Solar cells

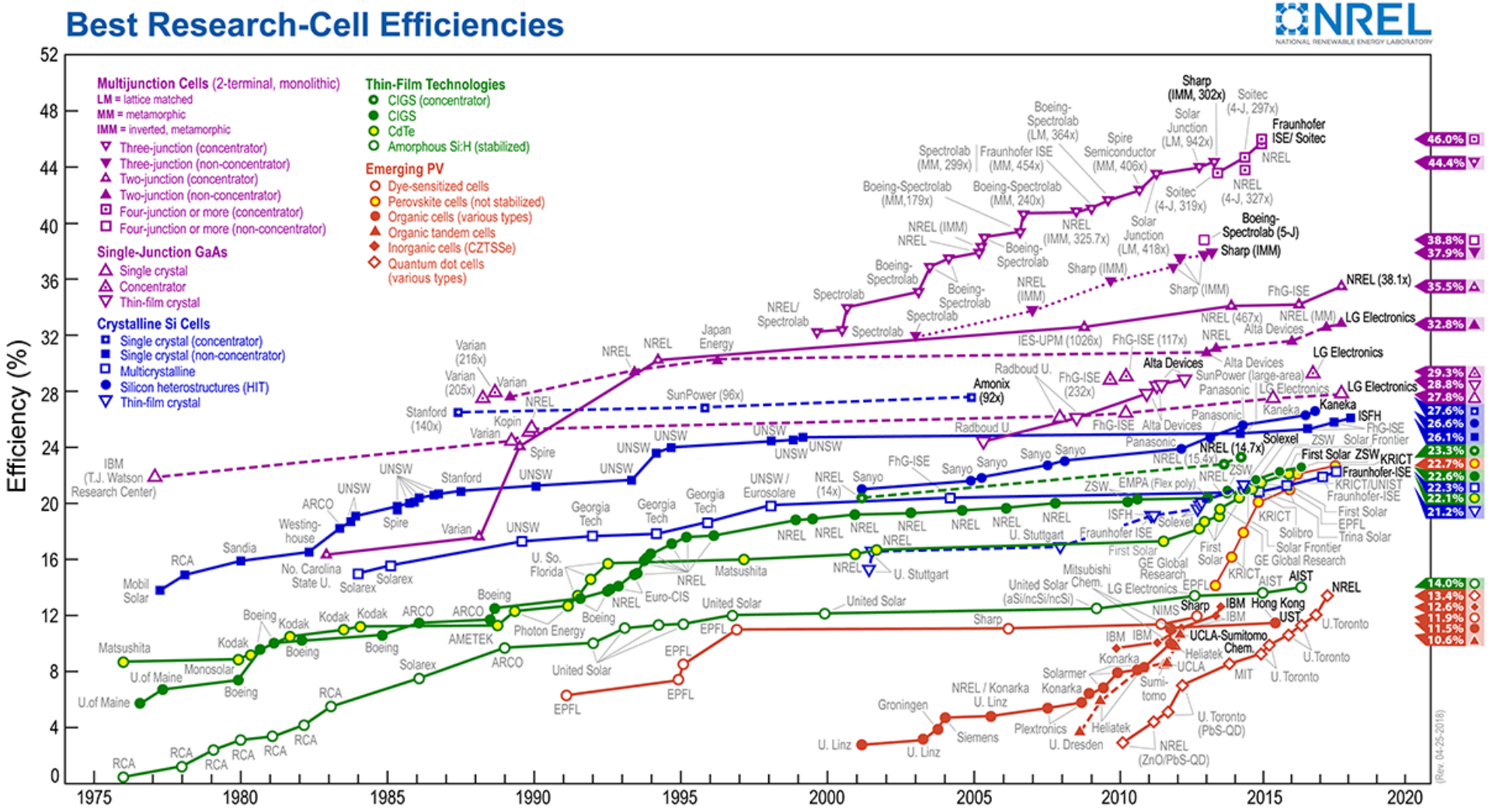
CEST CREI Research Thrust



www.pveducation.org

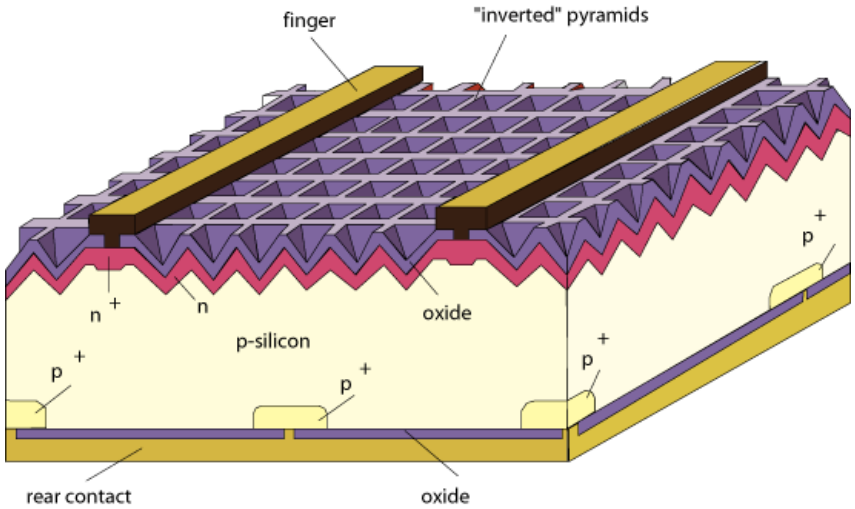
Understand → Optimize → Design → Manufacture

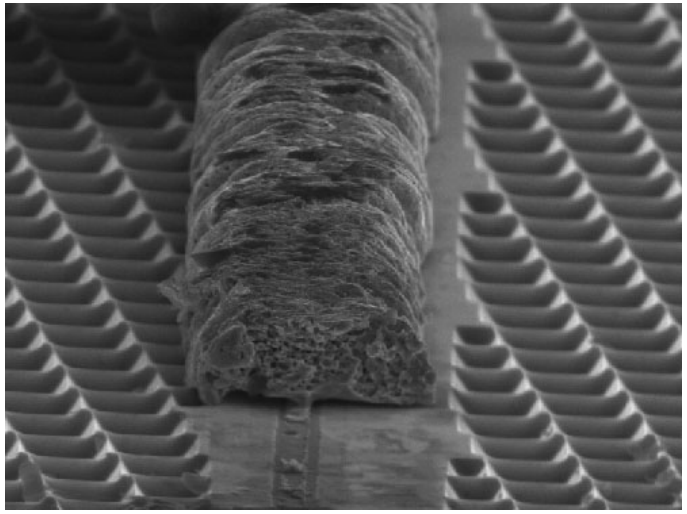
Power conversion efficiency



75-year evolution of Si solar cells: from 0 to 25%

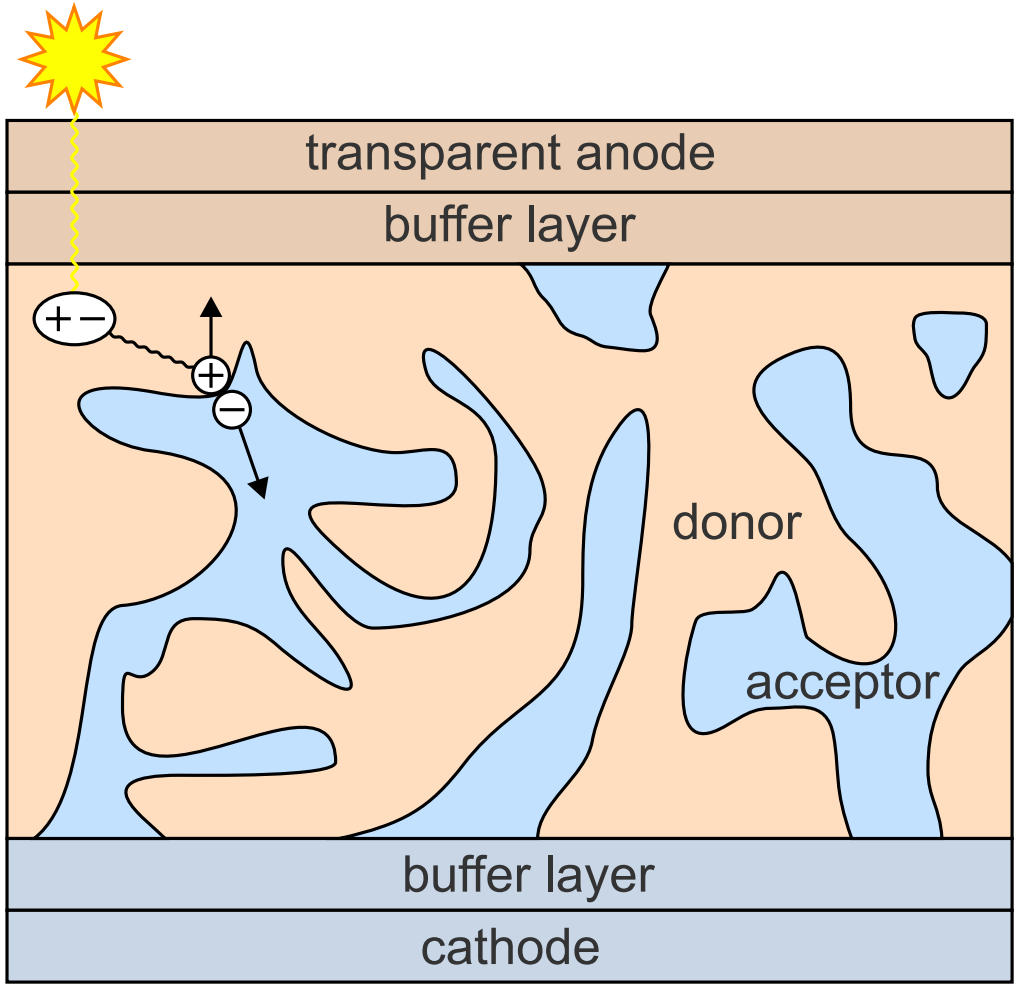
PERL – passivated emitter with rear locally diffused cell:





Reference: pveducation.org

Bulk-heterojunction solar cells



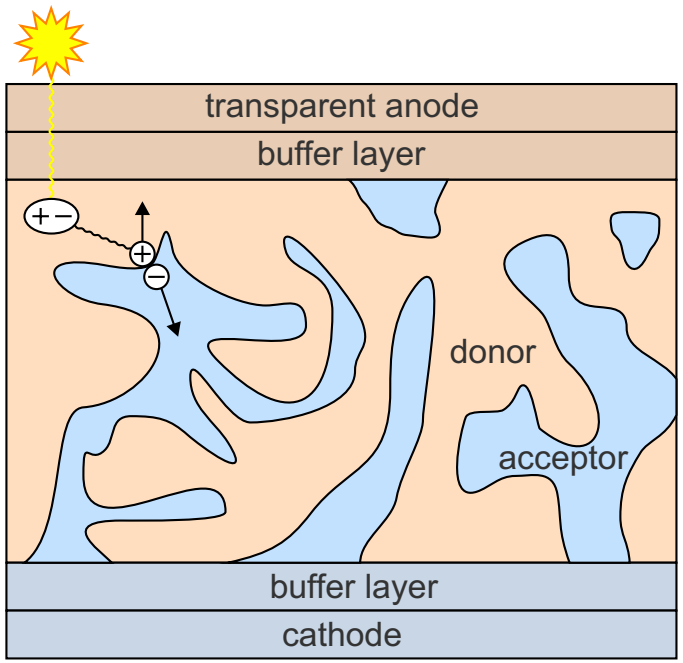
Optimize performance:

- Donor material
- Acceptor material
- Interface
- Morphology
- Contacts
- Light absorption
- Aging
- ...

$\Longrightarrow$ Structure & properties: 1) bulk material, 2) surface/interface

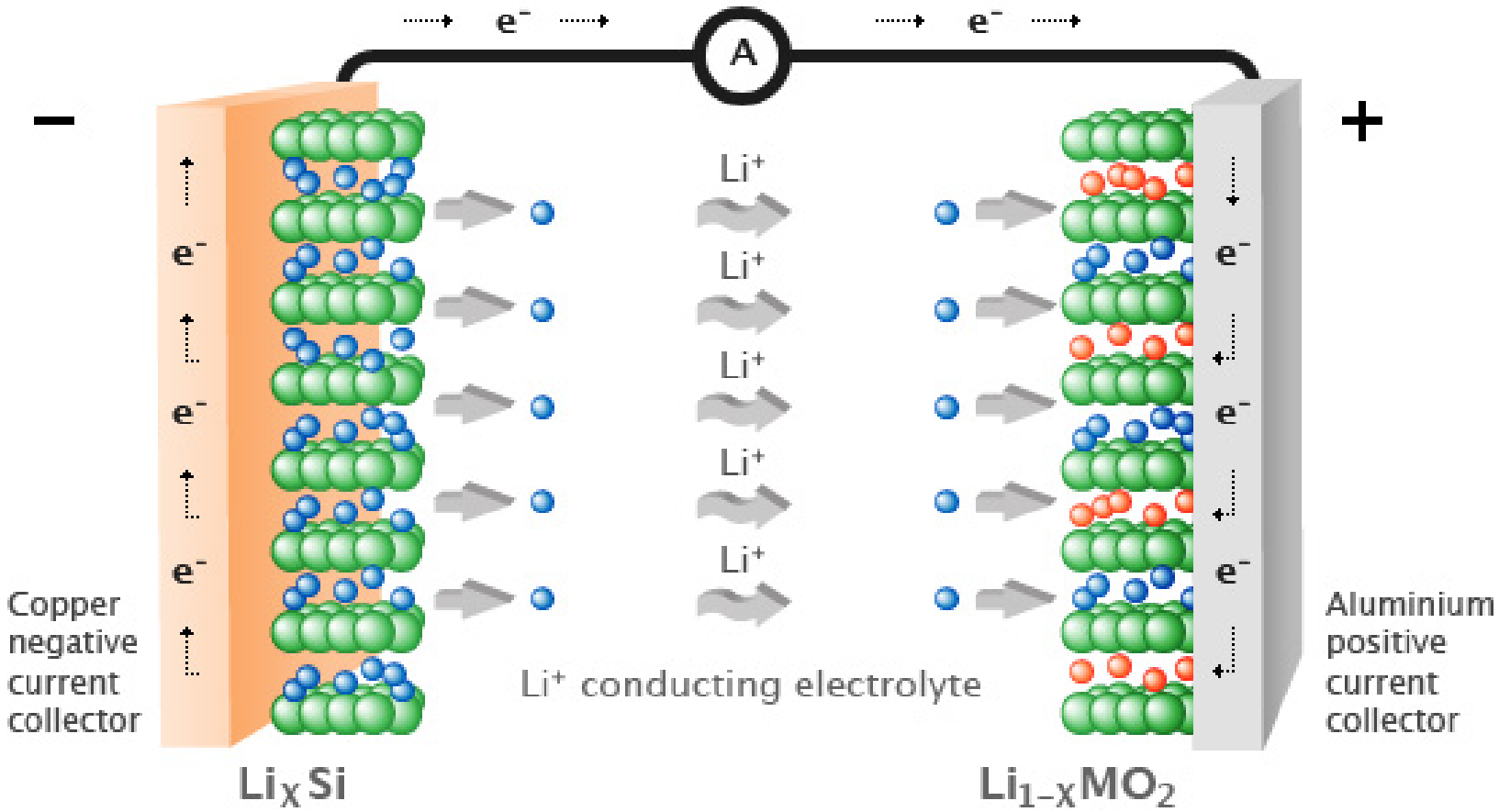
Properties to study and optimize

- Sunlight harvesting efficiency
- Exciton diffusion length
- Energy of charge carriers
- Mobility of charge carriers
- Efficiency of charge separation
- Degradation and aging



Case study 2: Metal-ion batteries

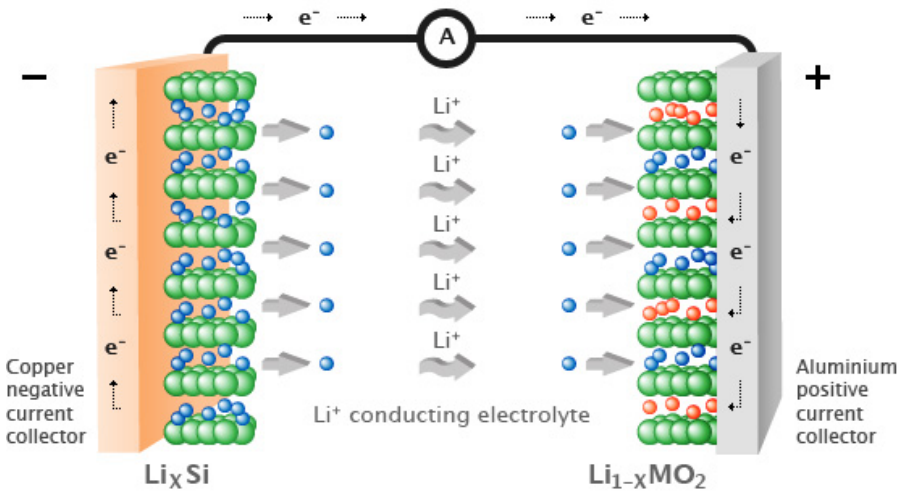
CEST CREI Research Thrust: two startup companies created – "Rustor" and "K-plus"



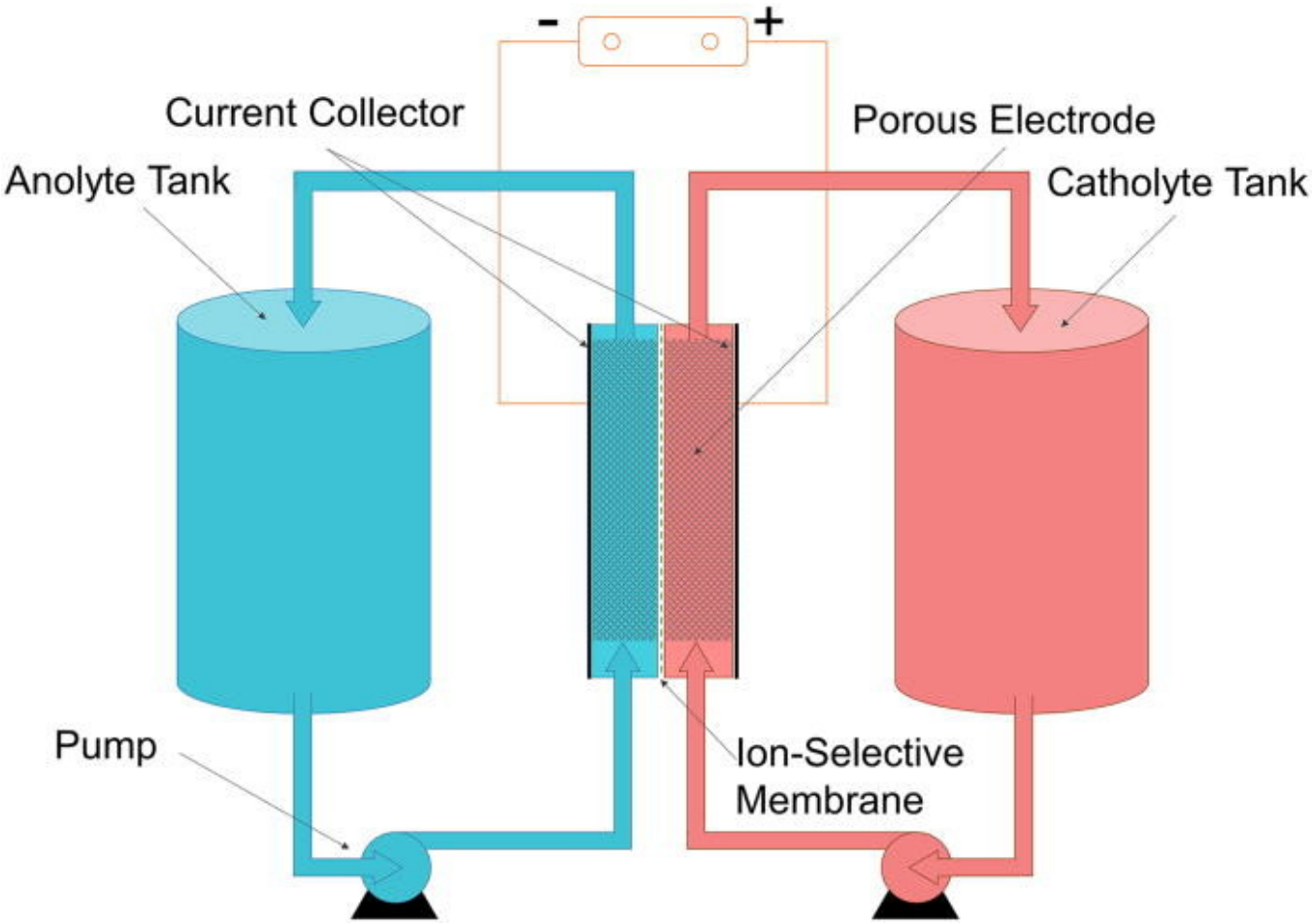
nexeon.co.uk

Properties to study and optimize

- Energy of charge carriers
- Diffusivity of charge carriers
- Cathode/anode capacity
- Charge/discharge reversibility
- Degradation and aging

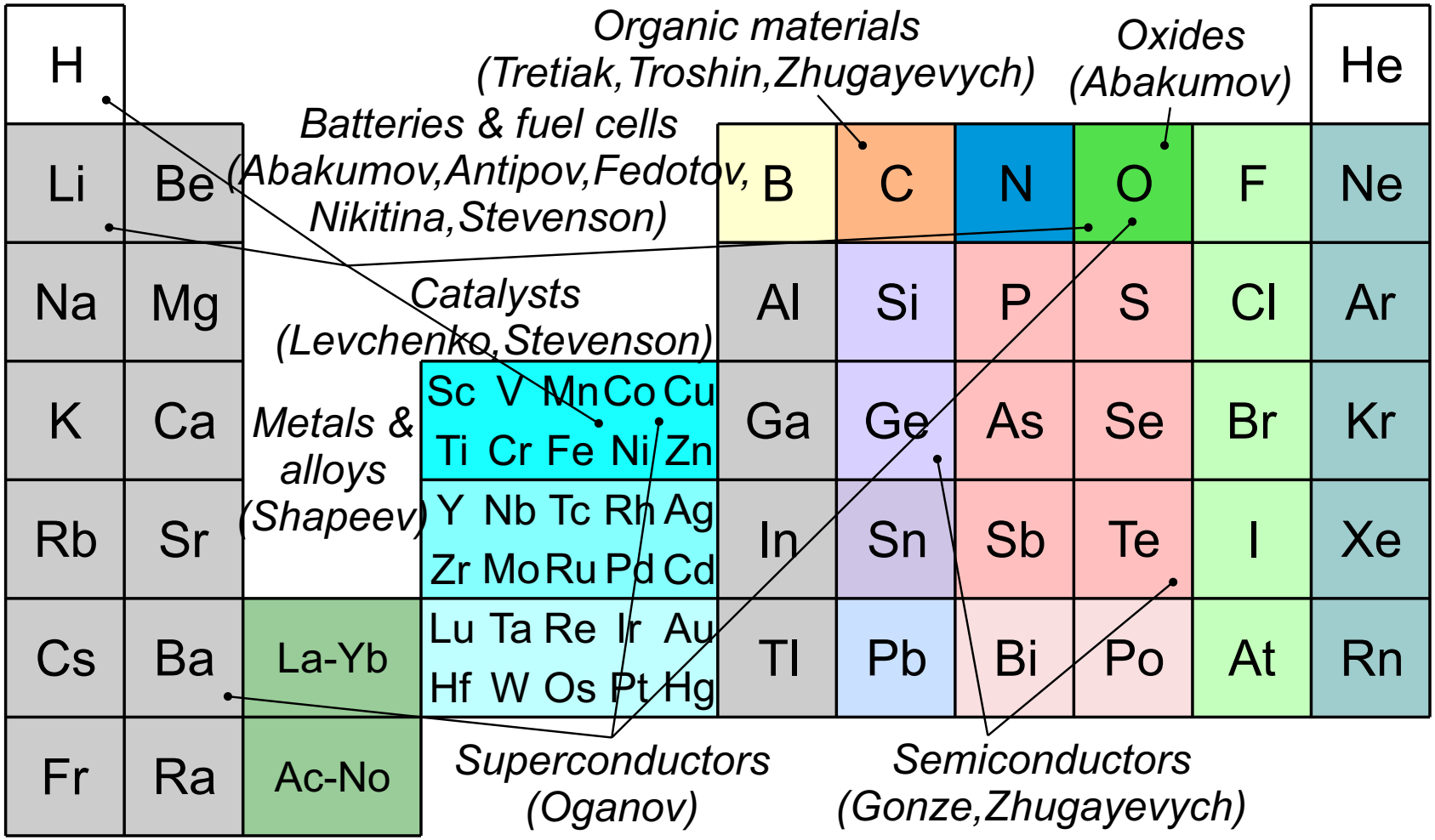


Case study 3: Redox flow batteries



The same concept but different device architecture $\implies$ different materials needed, different structure-property relationships

Part II overview: chemical composition perspective



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Part II overview: applications perspective at Skoltech

- Energy conversion and storage (CEST CREI):
  materials for energy generation, conversion, storage
  Abakumov, Antipov, Fedotov, Levchenko, Nikitina, Stevenson, Tretiak, Troshin, Zhugayevych
- Photonics and quantum materials (CPQM CREI):
  materials for electronics, spintronics, photonics
  Nasibulin
- Design, manufacturing and materials (CDMM CREI): composites
- Hydrocarbon recovery (CHR CREI): hydrocarbons
- Other CREIs: materials and devices
- Other: *Buchachenko, Gonze, Korsunksy, Oganov, Shapeev*

Course logistics

- Course web-page
- Syllabus
- Schedule and timeline
- Required software
- Part I exam: theory and practice
- 40 hours per week:
  - 12 in class
  - 13 reading and self-study (!)
  - 15 homeworks and projects
  - see details here
- Online mode

Bookmark these pages: Schedule, Canvas, Course web-page

Learning outcomes explained

See Syllabus for complete list

- Level your background (not to replace background courses)
- Introduce Materials Science language and concepts
  (be able to understand Materials Science papers and talks)
- Be familiar with Materials Science research at Skoltech
- Know classes of materials studied at Skoltech
- Know state of the art in a specific research area
- Expand your set of practical skills in Materials Science
  (no labs or hands-on experinece in this course)
- Here you can find or start research project (literature review)

Homework

- Familiarize yourself with course website
- Read carefully Syllabus
- Select at least one textbook for home reading
- Install required software